

LG-CISH: Lossless Language-Guided Coverless Image Steganography via CLIP Semantic Hashing and Base- N Positional Coding

Ayushman Bhattacharya^{1*}, Shreya Dutta², Piyali Sarkar³, Soumyashree Biswas⁴,
Harsh Raj⁵, Dr. Abhrendu Bhattacharya^{6*}
, Dr. Bidisha Bhabani⁷

¹⁻⁶Department of Computer Science and Engineering, JIS University,
Kolkata, West Bengal, India

¹bhattacharyaa599@gmail.com, ORCID: 0009-0006-5960-6308

²shreyadutta9503@gmail.com, ORCID: 0009-0004-8719-9151

³piyaliam02@gmail.com, ORCID: 0009-0008-8617-2247

⁴soumya7biswas@gmail.com, ORCID: 0009-0006-8702-8138

⁵harsh.cs.dev@gmail.com, ORCID: 0009-0004-9025-8204

⁶abhrendu.bhattacharya@jisuniversity.ac.in, ORCID: 0009-0009-8549-1024

⁷bidisha.bhabani@jisuniversity.ac.in, ORCID: 0009-0009-8549-1024

Corresponding Authors: Ayushman Bhattacharya^{1*} (bhattacharyaa599@gmail.com) and Dr.
Abhrendu Bhattacharya^{6*} (abhrendu.bhattacharya@jisuniversity.ac.in)

Abstract

Embedding-based image steganography perturbs cover pixels and is therefore detectable by modern statistical and CNN steganalysers. We present LG-CISH, a fully *lossless coverless* scheme that hides a secret message in a sequence of *unmodified* natural images: the identity and order of the images carry the data. LG-CISH combines two ideas. First, each codebook image is represented by a normalised CLIP embedding, so recovery is a robust semantic nearest-neighbour lookup that survives JPEG, resizing, noise, and aggressive cropping—unlike perceptual hashing. Second, base- N positional coding treats the encrypted, integrity-protected payload as a single integer written in radix N , giving the information-theoretic $\log_2 N$ bits per image. A thin zlib/AES-256/CRC-32 layer adds compression, confidentiality, and tamper detection, and two optional cover-shaping layers—a distinct-image *permutation* mode and an LLM-guided *alias* layer—reshape the cover without altering the encoded bits. On a 40-image codebook drawn from the UCID, Kodak, and USC-SIPI benchmark suites (5.322 bits/image), LG-CISH achieves bit-exact reconstruction (BER = 0), 100% accuracy across twenty-one channel attacks (JPEG down to quality 20, noise, downscaling to 25%, centre-cropping to 60%, and WebP conversion), chance-level steganalysis detectability (47.5% vs. 97.5% for LSB), and a statistically significant geometric-robustness advantage over perceptual hashing (paired Wilcoxon signed-rank, $p = 7.8 \times 10^{-3}$). We further quantify cover plausibility with an LLM judge, and analyse the capacity/undetectability and theme/plausibility trade-offs and the database-bound capacity limit of the coverless paradigm.

Keywords: coverless steganography; information hiding; CLIP; semantic hashing; base- N coding; robustness; steganalysis resistance.

1 Introduction

Steganography is the practice of hiding the very *existence* of a communication, in contrast to cryptography, which hides only its *content* [1]. The canonical setting is Simmons’ prisoner’s problem [2]: two parties wish to coordinate an escape while every message they exchange is inspected by a warden, who releases a message only if it appears innocuous. A message is therefore safe not when it cannot be read, but when its carrier cannot be distinguished from ordinary traffic. This distinction motivates everything that follows, and it is made precise in this section using the language of probability and information theory. The notation used throughout the paper is collected in Table 1.

It is worth stating at the outset why the problem is treated distributionally rather than perceptually. A cover image that is altered only slightly may look identical to a human observer and yet be flagged with near certainty by a machine detector, because the detector reads statistics that the eye ignores. Conversely, an image that is transmitted without any alteration cannot be flagged by *any* detector on distributional grounds, however powerful, because it is literally a sample of the same population as innocent images. The method proposed in this paper is built entirely around this second observation, and the purpose of the present section is to explain, with the necessary mathematics, why the observation is decisive and what must be engineered so that it can be exploited in practice.

Table 1: Principal notation used throughout the paper.

Symbol	Meaning
$\mathcal{I}$	space of images (e.g. 512×512 RGB)
P_C, P_S	probability laws of cover and stego objects on $\mathcal{I}$
$m, \hat{m}$	secret message and its reconstruction
k, κ	steganographic / AES-256 key
$\mathcal{C} = \{c_0, \dots, c_{N-1}\}$	shared codebook of N distinct images
$\Phi, e(\cdot)$	CLIP image encoder and its ℓ_2 -normalised embedding
$s(a, b)$	cosine similarity of two embeddings
γ_t	per-image decoding margin (top-1 minus top-2 similarity)
$T(\cdot)$	lossy transport channel (JPEG, resize, noise, crop)
L, B	sequence length and permutation block size
$P(N, B)$	number of ordered B -image blocks, $N!/(N - B)!$
$D(\cdot \ \cdot)$	Kullback–Leibler divergence

1.1 The covert-communication model

Let a cover object be a realisation of a random image $X \in \mathcal{I}$ drawn from the natural-image law P_C . A classical (embedding) stegosystem is a triple of maps: an embedding function that produces a stego object,

$$Y = \text{Emb}(X, m, k) \in \mathcal{I}, \quad (1)$$

an extraction function $\hat{m} = \text{Ext}(Y, k)$, and a shared key k . The warden observes the transmitted object and applies a binary detector $f : \mathcal{I} \rightarrow \{\text{cover}, \text{stego}\}$. The security of the scheme is governed entirely by how far the law P_S of the transmitted object Y departs from the innocent law P_C . This warden channel is illustrated in Figure 1.

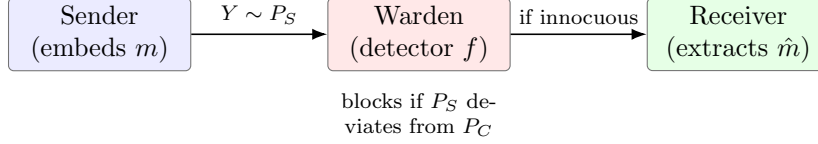


Figure 1: The prisoner’s problem. The transmitted object passes only if the warden’s detector cannot separate its law P_S from the natural-image law P_C . Security is a statement about distributions, not about pixels.

1.2 Threat model and assumptions

The setting adopted in this paper is made explicit through the following assumptions, which are standard for coverless steganography and are respected by the construction of Section 3.

Assumption 1.1 (Kerckhoffs’ principle). The warden is assumed to know the design of the stegosystem in full, including the coding, the feature map, and the cryptographic layer. Security is required to rest only on the secret quantities — the shared image database, the codebook ordering, and the AES key — and never on the obscurity of the algorithm.

Assumption 1.2 (Passive, computationally bounded warden). The warden is passive: transmitted objects are inspected and either forwarded unchanged or blocked, but they are not maliciously rewritten. The warden may run any efficient detector, including learned steganalysers, but cannot break AES-256.

Assumption 1.3 (Shared secret database). The sender and the receiver hold, offline, an identical and secret image database together with the same CLIP model. The database is treated as a shared key: an adversary without it cannot map images to indices. Loss of database secrecy reduces the scheme to its cryptographic layer only.

Assumption 1.4 (Benign but lossy channel). The transport channel T degrades images through content-preserving operations — re-compression, resizing, mild noise, and moderate cropping — but does not deliberately destroy them. The degradation is bounded so that semantic content is retained, which is the regime in which Equation (18) can be guaranteed.

Under Assumptions 1.1–1.4 the two objectives of the scheme can be separated cleanly: *undetectability* is an assertion about the marginal law of each transmitted image (§1.4–1.6), whereas *reliability* is an assertion about the recovery of indices from a degraded channel (§1.8). These are treated independently in the analysis that follows.

1.3 Embedding distortion and the payload it buys

Any embedding of the form of Equation (1) changes at least one pixel, so a distortion is incurred between the cover X and the stego Y . For n -pixel images the mean squared error and the peak signal-to-noise ratio are

$$\text{MSE}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2, \quad (2)$$

$$\text{PSNR}(X, Y) = 10 \log_{10} \frac{V_{\max}^2}{\text{MSE}(X, Y)} \text{ [dB]}, \quad (3)$$

Equations (2) and (3) define the distortion and its logarithmic quality scale, respectively, with $V_{\max} = 255$ for 8-bit images. Perceptual quality is frequently summarised by the structural similarity index

$$\text{SSIM}(X, Y) = \frac{(2\mu_X\mu_Y + C_1)(2\sigma_{XY} + C_2)}{(\mu_X^2 + \mu_Y^2 + C_1)(\sigma_X^2 + \sigma_Y^2 + C_2)}, \quad (4)$$

where μ, σ denote local means, variances, and covariance, and C_1, C_2 are stabilising constants; Equation (4) therefore complements PSNR with a structure-aware quality measure. The amount of secret data carried is measured by the relative payload (embedding rate)

$$\alpha = \frac{|m|}{n} \quad [\text{bits per pixel}]. \quad (5)$$

Equation (5) normalises the payload by image size. It is intuitively clear, and made precise below, that larger α forces larger distortion and hence a larger statistical footprint. This tension is the central obstacle that the present work seeks to avoid rather than merely to balance.

The trade-off is formalised by rate–distortion theory. For a given cover source there exists a capacity–distortion function $C(D)$, the largest secure payload achievable within a distortion budget D , which is non-decreasing and concave; a family of embedding methods differs only in how closely it approaches this frontier. Classical spatial and transform-domain schemes — least-significant-bit substitution [3], and DCT- or DWT-based embedding [4] — all operate strictly inside the region $D > 0$, and thus purchase every additional bit with additional distortion and, through Equation (9), with additional detectability. The proposal of this paper is to leave that frontier altogether by operating at the single point $D = 0$, which the coverless principle of §1.6 shows to be attainable.

1.4 Statistical detectability and its fundamental limits

The proper measure of steganographic security is not distortion but statistical distinguishability. Following Cachin [5], the (in)security of a stegosystem is quantified by the Kullback–Leibler divergence between the innocent and the transmitted laws,

$$D(P_C \parallel P_S) = \sum_{z \in \mathcal{I}} P_C(z) \log \frac{P_C(z)}{P_S(z)}, \quad (6)$$

and a scheme is called ε -secure when

$$D(P_C \parallel P_S) \leq \varepsilon, \quad \text{with perfect security at } \varepsilon = 0. \quad (7)$$

The security criterion in Equation (7) makes perfect security the zero-divergence case. The operational meaning of Equation (6) is exposed by hypothesis testing. For any warden detector f with false-alarm probability P_{FA} and missed-detection probability P_{MD} , the average error under equal priors is

$$P_E = \frac{1}{2}(P_{\text{FA}} + P_{\text{MD}}), \quad (8)$$

The detector error in Equation (8) is bounded below, through Pinsker’s inequality, by the divergence in Equation (6):

$$P_E \geq \frac{1}{2} - \sqrt{\frac{1}{2} D(P_C \parallel P_S)}. \quad (9)$$

Two consequences are read directly from Equation (9). First, when $D(P_C \parallel P_S) \rightarrow 0$ the best possible detector is driven to $P_E \rightarrow \frac{1}{2}$, i.e. to blind guessing. Second, any embedding that perturbs

the cover makes $P_S \neq P_C$ and therefore makes $D > 0$, which in turn caps how undetectable the scheme can be. A further limit, the square-root law of steganographic capacity [6], states that the largest payload that keeps D bounded grows only as the square root of the cover size,

$$|m^*| = \Theta(\sqrt{n}), \quad (10)$$

so that the *secure* embedding rate $\alpha^* = |m^*|/n = \Theta(1/\sqrt{n})$ actually vanishes as images grow. Modern convolutional steganalysers such as XuNet [7] and SRNet [8] approach these theoretical limits in practice, detecting embedding at rates once believed safe. Embedding steganography is thus squeezed from both sides: distortion is unavoidable (Equation (3)), and even small distortion is increasingly detectable (Equations (9)–(10)).

1.5 Detector performance and the cost of a single error

Two further quantities are used later and are introduced here for completeness. The discriminating power of a warden’s soft detector, which outputs a score whose mean shifts by $\Delta\mu$ between cover and stego while retaining variance σ^2 , is often summarised by the deflection coefficient

$$d^2 = \frac{(\Delta\mu)^2}{\sigma^2}, \quad (11)$$

According to Equation (11), a larger d^2 means an easier detection problem; for an unmodified transmitted object $\Delta\mu = 0$ and hence $d^2 = 0$. The overall quality of a detector across all thresholds is reported by the area under its receiver-operating-characteristic curve, $\text{AUC} \in [\frac{1}{2}, 1]$, with $\text{AUC} = \frac{1}{2}$ corresponding to the chance line of Equation (15).

On the receiver’s side, a complementary bound explains why losslessness is demanded rather than merely desired. Regarding the recovered index sequence as the output of a noisy channel whose input is the message, Fano’s inequality relates the probability of a decoding error P_e to the residual uncertainty about the message,

$$H(m \mid \hat{m}) \leq H_b(P_e) + P_e \log_2(|\mathcal{M}| - 1), \quad (12)$$

where $H_b(\cdot)$ is the binary entropy function and $\mathcal{M}$ the message alphabet. Moreover, because decoding is a deterministic function of the received images, the data-processing inequality guarantees that no post-processing can increase the information already present,

$$I(m; \hat{m}) \leq I(m; \{\tilde{y}_t\}). \quad (13)$$

Equations (12)–(13) show that a single mis-recovered index injects irreducible uncertainty that later stages cannot repair; this is precisely why the design in Section 3 aims for $P_e = 0$ on a clean channel and, where the channel is too harsh, converts an error into a *detected* failure by the integrity check rather than allowing silent corruption.

1.6 The coverless principle: removing the footprint

The escape from this squeeze is to *not embed at all*. In coverless steganography no cover is modified; instead, an ordinary image is *selected* (or generated) so that some intrinsic property of the image already encodes the desired bits [9, 10]. The transmitted object is then a genuine sample of the natural-image law, which is the key distributional fact:

$$Y \in \text{supp}(P_C) \text{ unmodified} \implies P_S = P_C \implies D(P_C \| P_S) = 0. \quad (14)$$

Substituting Equation (14) into Equation (9) yields the central promise of the paradigm,

$$P_E \geq \frac{1}{2}, \quad (15)$$

that is, undetectability holds *by construction* against any pixel-domain detector, because the transmitted images are, by definition, drawn from the same distribution as innocent images. The design problem is thereby transformed: the question is no longer “how may pixels be perturbed with least distortion?” but “how may a secret be *mapped* to a sequence of unmodified images, and *recovered* after that sequence has traversed a lossy channel?”. The three paradigms are contrasted in Table 2.

Within the coverless family two branches are distinguished. In the *generative* branch the transmitted image is synthesised from the secret by a generative model, so that new pixels are produced; although no cover is *edited*, the synthetic image may still deviate from the natural-image law and the mapping is not always exactly invertible. In the *mapping* (selection) branch, which is adopted here, the transmitted image is an existing, unmodified database image chosen so that its index encodes the bits; the transmitted object is then a genuine natural image and Equation (14) holds exactly. The mapping branch has historically relied on robust perceptual hashing to index the database, which is discriminative but fragile under geometric transport; the substitution of a semantic hash for a perceptual one, developed in §1.9, is the change that makes the mapping branch reliable over real channels.

Table 2: Comparison of steganographic paradigms. The coverless mapping paradigm, adopted here, is the only one that is both distortion-free and lossless.

Property	Embedding (LSB/DCT)	Coverless (generative)	Coverless (mapping, ours)
Modifies pixels	Yes	synthesises new pixels	No
$D(P_C P_S)$	> 0	small–moderate	0
Distortion (PSNR)	finite	finite	∞
Lossless recovery	usually	often not	Yes
Raw capacity	high	moderate	database-bound

1.7 The mapping formulation

Let a shared *codebook* $\mathcal{C} = \{c_0, c_1, \dots, c_{N-1}\}$ of N mutually distinct images be fixed offline. A message is regarded as an integer and represented in a positional number system whose “digits” select codebook images. If the codebook is treated as a radix- N alphabet, then each transmitted image carries

$$C_{\text{img}} = \log_2 N \quad [\text{bits per image}], \quad (16)$$

so that a message requiring L images conveys $L \log_2 N$ bits in total. For the codebook used in this work, $N = 40$ gives $C_{\text{img}} = \log_2 40 \approx 5.322$ bits per image. The mapping is a bijection between integers and index sequences, which is what makes the scheme *lossless*: no information is discarded at any stage of the encoding.

Two properties of this construction deserve emphasis. First, the assignment of indices to images is itself part of the shared secret: an adversary who observes the sequence but does not hold the codebook cannot invert the map, so the codebook acts as a large key and contributes the factor $N!$ to the keyspace of Equation (28). Second, the map is *positional*, so that the same image placed in two different positions of the sequence contributes different amounts to the encoded integer, exactly as a digit contributes differently according to its place. It is therefore the *ordered* sequence, and

not the multiset of images, that carries the message; the order is not incidental packaging but the substance of the code.

1.8 The channel and the recovery condition

Real messaging and social platforms do not deliver images untouched; they re-compress, resize, and re-encode them. A photograph forwarded through a popular messaging application, for example, is routinely re-encoded at a lower JPEG quality and downsampled to a bounded resolution, while the same image posted to a social network may be cropped to a fixed aspect ratio and stripped of its metadata. Any scheme that hides information in fragile low-level statistics is destroyed by such ordinary handling, which is why robustness to these operations is treated here as a first-class requirement rather than as an afterthought. This handling is modelled by a transport operator $T : \mathcal{I} \rightarrow \mathcal{I}$, so that the receiver observes distorted images $\tilde{y}_t = T(y_t)$. The receiver must map each distorted image back to the index the sender intended. Writing $c_{j^\star}$ for the image actually sent at position t , the decoded index is chosen as the codebook entry most similar to the received image,

$$\hat{i}_t = \arg \max_{0 \leq j < N} s(e(\tilde{y}_t), e(c_j)), \quad (17)$$

where $e(\cdot)$ is a feature map (developed in §1.9) and $s(\cdot, \cdot)$ its similarity. Recovery at position t is correct precisely when the true image outscores every rival, which is captured by a positive *decoding margin*

$$\gamma_t = s(e(\tilde{y}_t), e(c_{j^\star})) - \max_{j \neq j^\star} s(e(\tilde{y}_t), e(c_j)) > 0. \quad (18)$$

If the feature map is stable under transport, in the sense that $\|e(T(x)) - e(x)\|_2 \leq \delta$ for all codebook images, and if the codebook is separated so that distinct images differ by at least 2δ in feature space, then Equation (18) is guaranteed to hold and every index is recovered exactly. The reliability of the whole scheme therefore rests on choosing a feature map that is *robust to transport* yet *discriminative between images*.

The consequence of Equation (18) for a whole message is multiplicative. If correct recovery at the L positions is treated as independent with per-image failure probability $p_t = \Pr[\gamma_t \leq 0]$, then the probability that the entire message is recovered without a single error is

$$\Pr[\text{message correct}] = \prod_{t=1}^L (1 - p_t) \approx \exp\left(-\sum_{t=1}^L p_t\right), \quad (19)$$

so that even a small per-image error rate is amplified by a long sequence. Two design responses follow, and both are adopted in Section 3. First, the feature map is chosen so that p_t is essentially zero across the operating range of the channel, which drives Equation (19) towards one. Second, for degradations beyond that range the integrity check of Equation (27) converts the residual failures into *detected* errors, so that a corrupted message is discarded and retransmitted rather than silently accepted.

1.9 Semantic hashing with a vision–language model

Perceptual hashes, built from low-level pixel or frequency statistics, satisfy discriminability but degrade quickly under geometric transport such as cropping. In this work the feature map is instead taken from CLIP, a vision–language model trained by contrastive learning on a very large

corpus of image–text pairs [11]. Only its image encoder $\Phi : \mathcal{I} \rightarrow \mathbb{R}^d$ ($d = 512$) is used, and each image is represented by the ℓ_2 -normalised embedding

$$e(x) = \frac{\Phi(x)}{\|\Phi(x)\|_2}, \quad \|e(x)\|_2 = 1, \quad (20)$$

so that the similarity in Equation (17) is the cosine (inner) product

$$s(a, b) = \langle a, b \rangle = a^\top b \in [-1, 1]. \quad (21)$$

Equation (21) is the similarity score used for semantic matching. Because CLIP is trained to place *semantically* similar images near one another, its embedding of a JPEG-compressed, resized, or partially cropped image stays close to the embedding of the original: the visual meaning survives even when the pixels do not. This property is exactly the transport stability required by §1.8, and it is the reason a semantic hash succeeds where a perceptual hash fails.

The robustness just described is not accidental but a direct consequence of the training objective. The encoder is trained to maximise the agreement between an image and its matching caption relative to mismatched captions, through a symmetric contrastive (InfoNCE) loss. For a batch of K image–text pairs with image embeddings v_i and text embeddings w_j and temperature ϑ , the image-to-text term of that loss is

$$\mathcal{L} = -\frac{1}{K} \sum_{i=1}^K \log \frac{\exp(\langle v_i, w_i \rangle / \vartheta)}{\sum_{j=1}^K \exp(\langle v_i, w_j \rangle / \vartheta)}, \quad (22)$$

with an analogous text-to-image term. Minimising Equation (22) over a very large and varied corpus forces the image encoder to place all views of the same underlying scene — including compressed, rescaled, or lightly cropped views — close together, because such views share the same caption. The embedding is therefore approximately invariant to precisely the content-preserving degradations that make up the transport channel of Assumption 1.4, which is the mechanism that keeps the decoding margin of Equation (18) positive in practice.

1.10 Capacity of the image-sequence channel

Two lossless codes are developed in this paper. The first, base- N positional coding, attains the per-image ceiling of Equation (16) but permits an image to repeat within a sequence. A repeated image can make a cover look unnatural, so a second code is introduced in which every block of B images consists of *distinct* images chosen without repetition. The number of such ordered blocks is the falling factorial

$$P(N, B) = \frac{N!}{(N - B)!} = \prod_{t=0}^{B-1} (N - t), \quad (23)$$

and a block of B images therefore carries $\log_2 P(N, B)$ bits, i.e.

$$C_{\text{perm}} = \frac{1}{B} \log_2 P(N, B) \quad [\text{bits per image}]. \quad (24)$$

Using Stirling’s approximation, $\log_2 N! \approx N \log_2(N/e)$, the full-block case $B = N$ satisfies $C_{\text{perm}} \approx \log_2(N/e)$, which is below the base- N ceiling of Equation (16) by roughly $\log_2 e \approx 1.443$ bits per image; smaller blocks recover most of the loss. Thus Equation (16) and Equation (24) bound a controllable trade-off between raw capacity and cover plausibility, which is one of the design levers of the method.

1.11 The lossless requirement and integrity

Since a single mis-recovered index corrupts every subsequent bit, the scheme is designed so that, on a clean channel, reconstruction is bit-exact. Denoting the original and recovered payload bits by p_b and $\hat{p}_b$ over B_{tot} bits, the bit error rate is

$$\text{BER} = \frac{1}{B_{\text{tot}}} \sum_{b=1}^{B_{\text{tot}}} \mathbb{1}[\hat{p}_b \neq p_b], \quad (25)$$

and, because every coding stage is a bijection, $\text{BER} = 0$ whenever all indices are recovered ($\gamma_t > 0 \forall t$, Equation (18)). For the lossless case the mutual information between the sent and the reconstructed message attains the message entropy,

$$I(m; \hat{m}) = H(m), \quad (26)$$

i.e. the covert channel is noiseless end-to-end. To detect the rare event of a transport failure, a 32-bit cyclic redundancy check is appended, which flags a corrupted message with probability

$$\Pr[\text{undetected error}] \approx 2^{-32}. \quad (27)$$

Finally, confidentiality and key strength are provided by AES-256-CBC, giving a combined secret space of

$$K = N! \cdot 2^{256}, \quad (28)$$

the product of the number of codebook orderings and the AES key space.

1.12 Source coding, expected length, and a worked example

The number of images that a message occupies is governed by information theory. If the message is modelled as a source with entropy $H(m)$ bits, then, by Shannon's source-coding theorem, no lossless compressor can represent it in fewer than $H(m)$ bits on average,

$$\mathbb{E}[|\text{ZLIB}(m)|] \geq \frac{H(m)}{8} \quad [\text{bytes}], \quad (29)$$

Equation (29) supplies the theoretical lower bound, and the compressor of Equation (32) is used precisely to push the encoded size towards this floor before any image is chosen. Once the framed payload occupies $|f|$ bytes, the base- N code of Equation (16) produces an expected sequence length

$$\mathbb{E}[L] \approx \frac{8|f|}{\log_2 N}, \quad (30)$$

so that capacity and length are reciprocal: a larger codebook shortens the cover. A concrete illustration is given in Table 3 for the codebook used in this work ($N = 40$, $\log_2 N \approx 5.322$ bits per image). It is emphasised that the figures in Table 3 are dominated, for short messages, by the fixed overhead of the cryptographic frame (the CRC, the AES initialisation vector and padding, and the compression header); the per-image rate of 5.322 bits is the asymptotic behaviour that governs long messages.

Table 3: Worked example of Equation (30) for the $N = 40$ codebook. The framed size includes compression, a 4-byte CRC, and AES padding; short messages are overhead-dominated, whereas long messages approach $8|f|/\log_2 40$ images.

Message class	Payload bits	Images $\approx 8 f /\log_2 40$	Rate regime
Short (≤ 50 chars)	~ 300	~ 57	overhead-dominated
Medium (50–200 chars)	~ 800	~ 150	transitional
Long (200–1000 chars)	~ 2100	~ 400	asymptotic (5.322 b/img)

1.13 Problem statement, design goals, and contributions

The preceding development frames the problem addressed in this paper. *Given* a shared image database and a shared vision–language model, a scheme is sought that (i) transmits only unmodified natural images, so that Equations (14)–(15) hold and detection is reduced to chance; (ii) is lossless, so that Equations (25)–(26) hold on a clean channel; and (iii) recovers correctly after realistic transport, so that Equation (18) is satisfied across compression, noise, resizing, and cropping. The following contributions are made:

1. A coverless mapping scheme, **LG-CISH**, is proposed in which the identity and order of unmodified images carry the secret, and reliable recovery is obtained by CLIP semantic hashing (§1.9, §3).
2. Two interchangeable lossless codes are formalised — base- N positional coding at the $\log_2 N$ ceiling (Equation (16)) and Lehmer-permutation coding with a no-repeat guarantee (Equations (23)–(24)) — together with a whitening step that removes an ordering artefact.
3. A confidentiality and integrity layer (AES-256-CBC, CRC-32) and an optional large-language-model-verified alias mechanism for cover plausibility are integrated without affecting losslessness (§3.1.2, §3.3).

1.14 Organisation of the paper

The remainder of the paper is organised as follows. Related work is reviewed in Section 2. The proposed method — notation and shared state, cryptographic framing, the two coding channels, and the encoding and decoding procedures — is developed in Section 3. The experimental setup, results, robustness, steganalysis resistance, ablation, comparison, and statistical validation are reported in the subsequent sections.

2 Literature Review

2.1 Traditional Image Steganography Techniques

Image steganography has been extensively investigated as a means of concealing secret information within digital images while maintaining visual imperceptibility. Conventional approaches can be broadly categorized into spatial-domain and transform-domain methods. Spatial-domain techniques, particularly Least Significant Bit (LSB) substitution, achieve high embedding capacity by directly modifying pixel values; however, they remain vulnerable to statistical steganalysis and image-processing attacks. Transform-domain approaches such as the Discrete Cosine Transform

(DCT), the Discrete Wavelet Transform (DWT), and hybrid DWT–DCT methods improve robustness by embedding information into frequency coefficients, thereby reducing perceptual distortion and increasing resistance to compression attacks [12, 13, 14, 15].

Kadhim et al. [16] conducted a comprehensive survey of image steganography and concluded that a trade-off always exists among payload capacity, imperceptibility, and robustness. Similarly, Subramanian et al. [12] highlighted that modern transform-domain schemes achieve superior image quality compared with spatial-domain approaches but still rely on pixel modification. Mandal et al. [17] further demonstrated that hybrid transform methods can improve robustness but remain susceptible to advanced detection mechanisms.

Despite continuous improvements in embedding strategies, traditional steganographic systems inherently modify image content, which leaves detectable statistical traces that can be exploited by steganalysis algorithms.

2.2 Steganalysis and Detection Challenges

The increasing sophistication of steganographic techniques has been accompanied by significant advances in steganalysis. Statistical detectors such as chi-square analysis [18], RS analysis [19], and histogram-based methods have proven effective in identifying hidden information embedded within digital images. Recent developments in machine learning and deep learning have further enhanced the detection capability of steganalysis systems.

Kombrink et al. [13] presented a detailed survey of modern image-steganography detection techniques and reported that convolutional neural network (CNN)-based detectors consistently outperform conventional statistical approaches. Kheddar et al. [20] reviewed deep-learning-based steganalysis frameworks and showed that deep neural architectures can successfully identify subtle embedding artifacts across diverse multimedia formats. Likewise, Apau et al. [21] observed that even highly optimized steganographic algorithms often exhibit detectable statistical inconsistencies under advanced analysis.

These findings indicate that, regardless of embedding domain, the fundamental weakness of conventional steganography arises from the modification of pixel values. Consequently, researchers have increasingly explored coverless communication paradigms that avoid image modification altogether.

2.3 Coverless Image Steganography

Coverless image steganography has emerged as a promising alternative to conventional data-hiding techniques. Unlike traditional methods, coverless approaches do not alter image pixels; instead, secret information is conveyed through image selection, retrieval, indexing, or semantic relationships among images.

Qin et al. [14] presented one of the most comprehensive surveys on coverless image steganography and classified existing approaches into image retrieval-based, image mapping-based, and semantic association-based methods. Their study demonstrated that coverless communication inherently avoids statistical detection because no image modifications are introduced during transmission. Meng et al. [22] further reviewed recent developments in coverless steganography and highlighted that image-database construction and semantic indexing remain critical challenges affecting practical deployment.

Although coverless methods significantly improve resistance to steganalysis, most existing approaches rely on handcrafted features, image-hashing techniques, or predefined mapping tables.

Such mechanisms often suffer from limited semantic representation capability and reduced scalability when handling complex textual information.

More recent coverless schemes have pursued robustness and capacity through neglected-dataset construction [23], end-to-end learned hash generation [24], SURF-based retrieval [25], and image-restoration front-ends [26], while a parallel generative branch synthesises stego images by image mapping and diffusion models [27, 28]; broad surveys catalogue this rapid growth [10, 29]. Deep image-hiding networks that embed one image within another [30, 31] remain, by contrast, pixel-modifying and thus detectable.

2.4 Semantic Representation and Hash-Based Retrieval

Recent advances in multimedia retrieval have demonstrated that semantic representation learning provides an effective mechanism for bridging heterogeneous data modalities through high-dimensional feature embeddings. Instead of relying solely on low-level visual descriptors, modern retrieval frameworks aim to preserve semantic relationships among data samples by learning discriminative latent representations that capture contextual and conceptual similarities. Such representations facilitate robust similarity matching and improve retrieval performance in large-scale image databases.

Hu et al. [32] proposed a Joint Semantic Preserving Sparse Hashing (JSPSH) framework for cross-modal retrieval, in which semantic consistency between different modalities is maintained during the hash-learning process. Their approach simultaneously exploits semantic labels and sparse-representation constraints to generate compact binary hash codes while preserving inter-modal semantic correlations. Experimental results demonstrated that semantic-preserving hashing significantly improves retrieval accuracy by ensuring that semantically related instances are mapped to neighboring regions in the Hamming space. The study further highlighted the importance of maintaining semantic structure during binary-code generation, thereby enabling efficient large-scale retrieval with reduced storage and computational overhead.

Similarly, Li et al. [15] introduced a Semantic Preservation-Based Hash Code Generation (SPHCG) framework for fine-grained image retrieval. Unlike conventional hashing methods that primarily focus on visual similarity, their approach emphasizes the preservation of subtle semantic distinctions among visually similar image categories. By integrating semantic information into the hash-learning process, the generated binary codes retain discriminative characteristics that improve retrieval precision in complex image datasets. Their findings demonstrate that semantic-aware hash generation can effectively enhance retrieval robustness while maintaining computational efficiency.

These studies collectively indicate that semantic-preserving hashing provides an effective mechanism for transforming high-dimensional semantic representations into compact binary codes while retaining meaningful neighbourhood relationships.

Although semantic-preserving hashing has achieved remarkable success in cross-modal retrieval and fine-grained image-retrieval applications [32, 15], its integration with coverless image steganography remains largely unexplored. Existing coverless communication schemes predominantly rely on perceptual similarity, image indexing, or handcrafted feature matching, without leveraging the semantic-understanding capabilities offered by modern representation-learning frameworks. Consequently, the potential of combining semantic embeddings, semantic hashing, and image retrieval for secure coverless communication has not yet been fully investigated. This limitation motivates the development of the proposed LLM-Guided Coverless Image Steganography using Semantic Hash Mapping (LG-CISH) framework, which exploits semantic-preserving hash representations to establish a robust mapping between textual semantics and image-based communication channels.

2.5 Emerging Role of Large Language Models in Secure Communication

The rapid advancement of Large Language Models (LLMs) has introduced new possibilities for semantic communication systems. LLMs possess the capability to transform textual information into semantically meaningful token sequences and contextual embeddings that preserve linguistic relationships. Such representations provide a natural bridge between human language and machine-understandable semantic spaces.

Recent studies in semantic communications suggest that transmitting meaning rather than raw bits can improve communication efficiency and robustness. However, the application of LLM-generated semantic embeddings to coverless steganography remains largely unexplored. Existing steganographic research primarily focuses on payload optimization, distortion minimization, and robustness enhancement, while semantic understanding is rarely incorporated into the communication pipeline.

Consequently, there exists a significant opportunity to leverage LLM-driven semantic encoding for secure image-based communication systems.

2.6 Research Gap and Motivation

The literature reveals several important limitations in existing steganographic and coverless communication approaches, as summarised in Table 4.

Table 4: Comparison of existing methodology, their limitations, and gap analysis.

References	Method	Limitation	Research Gap
Kadhim et al. [16]	Spatial and transform-domain image steganography	Focuses on pixel-modification techniques and trade-offs among capacity, robustness and imperceptibility.	Need a pixel-modification-free (coverless) communication framework that completely avoids embedding artifacts.
Subramanian et al. [12]	Transform-domain (DCT, DWT, hybrid)	Improves image quality but still modifies transform coefficients.	Develop a framework eliminating coefficient modification while preserving security through semantic image selection.
Mandal et al. [17]	Hybrid transform-domain steganography	Vulnerable to advanced statistical and deep-learning steganalysis.	Design a steganalysis-resistant coverless communication system without altering image statistics.
Kombrink et al. [13]	CNN-based steganalysis survey	Demonstrates high detection accuracy against conventional steganography.	Develop communication methods that generate no embedding artifacts detectable by CNN-based steganalysis.
Kheddar et al. [20]	Deep-learning-based multimedia steganalysis	Detects subtle embedding traces even in optimized methods.	Introduce a communication paradigm based on semantic retrieval rather than embedding.
Apau et al. [21]	Statistical analysis of modern steganography	Hidden data still leaves statistical inconsistencies.	Remove statistical footprints through coverless semantic communication.
Qin et al. [14]	Coverless image steganography	Mainly relies on image retrieval, mapping and handcrafted descriptors.	Replace handcrafted descriptors with deep semantic embeddings and semantic hashing for robust retrieval.
Meng et al. [22]	Coverless image steganography	Database construction and semantic indexing remain challenging.	Develop an automatic semantic hash-mapping mechanism for scalable image retrieval.
Hu et al. [32]	Joint Semantic Preserving Sparse Hashing (JSPSH)	Designed for cross-modal retrieval, not secure communication.	Integrate semantic-preserving hashing into coverless image steganography for secure message transmission.
Li et al. [15]	Semantic Preservation-Based Hash Code Generation (SPHCG)	Focuses on fine-grained image retrieval only.	Extend semantic hashing to coverless steganographic communication and message reconstruction.

Conventional steganography techniques modify image pixels and therefore remain vulnerable to statistical and deep-learning-based steganalysis [17, 22, 32]. Existing coverless image-steganography approaches largely depend on handcrafted image descriptors, image-retrieval mechanisms, or predefined mapping rules that provide limited semantic expressiveness [16, 20]. Current image retrieval-based systems generally focus on visual similarity rather than semantic understanding, restricting their ability to encode complex natural-language messages. Existing studies have not adequately explored the integration of Large Language Models with semantic hashing for coverless communication, and a unified framework that combines semantic-token generation, semantic embedding,

semantic hashing, perceptual image hashing, image retrieval, and semantic reconstruction remains absent from the literature.

Motivated by these challenges, this work proposes a novel LLM-Guided Coverless Image Steganography using Semantic Hash Mapping (LG-CISH) framework. The proposed method introduces an end-to-end semantic communication architecture in which a secret message is transformed into semantic tokens, encoded into semantic hashes, matched against perceptual image hashes stored in an image database, and reconstructed through reverse semantic mapping. By eliminating pixel modification entirely, the proposed framework aims to achieve high steganalysis resistance while simultaneously exploiting the semantic-reasoning capabilities of modern Large Language Models.

2.7 Research Objectives

Following directly from the motivation in Section 2.6, the research objectives are to:

1. design a coverless communication mechanism that carries information through the identity and order of unmodified natural images rather than through pixel or transform-coefficient modification;
2. construct the Architecture Model of Section 3.1, in which cryptographic framing, lossless positional coding, semantic image retrieval, and exact decoding form a single reproducible pipeline;
3. use CLIP semantic embeddings to maintain reliable image-index recovery after content-preserving channel operations such as JPEG compression, resizing, noise, and cropping; and
4. evaluate the resulting system in terms of losslessness, payload capacity, channel robustness, steganalysis resistance, and cover plausibility.

3 Proposed Method

3.1 Architecture Model

The proposed method is a lossless, coverless pipeline in which a secret message is carried by an ordered sequence of unmodified codebook images. The sender and the receiver are assumed to share, offline, three artefacts: an image database, the CLIP image encoder, and the resulting codebook that assigns a unique index to each database image. On the sender side the message is first compressed, integrity tagged, and optionally encrypted, then converted into a sequence of codebook indices by one of two lossless codes, and finally emitted as the corresponding images. The images traverse an arbitrary lossy channel. On the receiver side each received image is embedded with CLIP and matched to the nearest codebook entry to recover its index, after which the coding and cryptographic steps are inverted to reconstruct the message exactly. This Architecture Model is illustrated in Figure 2 and is referred to throughout the encoding, decoding, and evaluation stages. Each component is developed in turn in the following subsections.

The architecture is organised around a deliberate separation of concerns, each concern being handled by a distinct stage so that the stages can be reasoned about and modified independently. *Confidentiality and integrity* are handled first, by the cryptographic frame of §3.1.2, so that everything downstream operates on an opaque, integrity-protected byte string and the coding layer need know nothing about the message semantics. *Capacity and cover plausibility* are handled next, by the choice between the two coding channels of §3.2, which converts the byte string into indices

at a controllable rate and with an optional no-repeat guarantee. *Undetectability* is inherited for free from the coverless principle, because the emitted objects are unmodified database images. Finally, *reliability* is delegated entirely to the semantic feature map at decode time, so that the sender performs no neural computation at all and the burden of channel robustness falls on a single, well-understood component.

The ordering of the pipeline is not arbitrary. Compression is applied before encryption because encrypted data is effectively incompressible, so any compression must precede the cipher; the integrity tag is computed over the compressed payload so that a corruption anywhere in transit is caught before decompression is attempted; and the positional coding is applied last, on the fully framed bytes, so that it is agnostic to whether encryption was used. This layered discipline is what allows the losslessness argument of Proposition 3.1 to be stated as a simple composition of invertible stages.

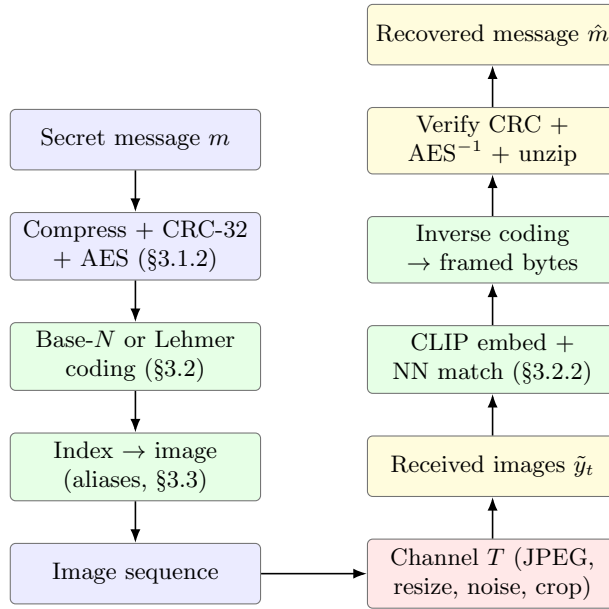


Figure 2: End-to-end flowchart of LG-CISH. The left column is the sender, the right column the receiver; the two are joined by the lossy channel T . Every transformation is deterministic and invertible, so the covert channel is lossless whenever the indices are recovered correctly.

3.1.1 Pre-processing: notation and shared state

The shared codebook is denoted $\mathcal{C} = \{c_0, \dots, c_{N-1}\}$, and its CLIP embeddings are stored as the rows of a matrix $\mathbf{C} \in \mathbb{R}^{N \times d}$ with $\mathbf{C}_j = e(c_j)$, where $e(\cdot)$ is the normalised embedding of Equation (20). So that the nearest-neighbour rule of Equation (17) is unambiguous, the codebook is required to be *separated*: no two distinct images may exceed a similarity threshold τ ,

$$\max_{i \neq j} s(\mathbf{C}_i, \mathbf{C}_j) \leq \tau, \quad (\tau = 0.85). \quad (31)$$

A candidate database is reduced to a separated codebook by the greedy construction of Algorithm 1, which retains an image only if it is not too similar to any image already kept.

The size N of the codebook is a design parameter that trades capacity against the difficulty of maintaining separation. By Equation (16) the per-image capacity grows only logarithmically in N , so doubling the codebook adds just one bit per image; yet as N grows it becomes progressively

Algorithm 1 Codebook construction (pre-processing)

```
1: Input: candidate image set  $\mathcal{D}$ , encoder  $\Phi$ , threshold  $\tau$ 
2: Output: codebook embeddings  $\mathbf{C}$ , image paths, size  $N$ 
3:  $E \leftarrow [e(x) \text{ for } x \in \mathcal{D}]$   $\triangleright$  normalise, Equation (20)
4:  $\mathcal{K} \leftarrow \emptyset$ 
5: for  $j = 0$  to  $|\mathcal{D}| - 1$  do
6:   if  $\max_{i \in \mathcal{K}} s(E_i, E_j) \leq \tau$  then  $\mathcal{K} \leftarrow \mathcal{K} \cup \{j\}$   $\triangleright$  Equation (31)
7: end for
8: return  $\mathbf{C} \leftarrow E[\mathcal{K}]$ ,  $\text{paths}[\mathcal{K}]$ ,  $N \leftarrow |\mathcal{K}|$ 
```

Algorithm 2 Cryptographic framing (WRAP / UNWRAP, AES-256-CBC)

```
1: function WRAP( $m, \kappa$ )
2:    $p_0 \leftarrow \text{ZLIB}(\text{UTF8}(m))$   $\triangleright$  Equation (32)
3:    $t \leftarrow p_0 \parallel \text{CRC32}(p_0)$   $\triangleright$  append 4-byte CRC
4:   if  $\kappa$  given then  $IV \leftarrow \text{RAND}(16)$ ; return  $IV \parallel \text{AES-CBC}_\kappa(\text{PAD}(t), IV)$ 
5:   else return  $t$   $\triangleright$  CRC-only mode
6: function UNWRAP( $f, \kappa$ )
7:   if  $\kappa$  given then  $t \leftarrow \text{UNPAD}(\text{AES-CBC}_\kappa^{-1}(f_{[16:]}, f_{[16:]}))$  else  $t \leftarrow f$ 
8:   split  $t = p_0 \parallel c$ ; if  $c \neq \text{CRC32}(p_0)$  then reject
9:   return  $p_0$ 
```

harder to keep every pair of images below the threshold τ of Equation (31), because the embedding space becomes more densely populated and the minimum pairwise separation shrinks. A moderate codebook is therefore preferred: it keeps the separation margin $1 - \max_{i \neq j} s(\mathbf{C}_i, \mathbf{C}_j)$ comfortably positive, which is exactly the quantity that, together with the transport stability of Assumption 1.4, guarantees correct recovery through Definition 3.1. The threshold τ is thus best read not as an arbitrary constant but as the knob that sets how much channel degradation the codebook is provisioned to absorb.

Definition 3.1 (Separated codebook). A codebook $\mathcal{C}$ is τ -separated if every pair of its embeddings satisfies Equation (31). Under τ -separation and transport stability $\|e(T(c_j)) - e(c_j)\|_2 \leq \delta$ with $2\delta < 1 - \tau$, the recovery rule of Equation (17) returns the correct index for every codebook image.

3.1.2 Cryptographic framing

Before any image is chosen, the message is wrapped in a thin layer that provides compression, integrity, and confidentiality. The message is first losslessly compressed with ZLIB, which shortens the sequence; a 32-bit CRC is then appended for integrity; and the result is optionally encrypted with AES-256-CBC [33] under a 256-bit key κ , with a random 16-byte initialisation vector prepended to the ciphertext. Writing these operations explicitly, the framed payload is

$$p_0 = \text{ZLIB}(\text{UTF8}(m)), \quad (32)$$

$$f = \text{AES}_\kappa(p_0 \parallel \text{CRC32}(p_0)), \quad (33)$$

where the AES step reduces to the identity when no key is supplied, in which case integrity is still provided by the CRC. The encryption itself follows the cipher-block-chaining recurrence: with the

Algorithm 3 Lehmer block coding (PERMEncode / PERMDecode)

```

1: function PERMEncode( $u, N, B$ )  $\triangleright$  integer  $\rightarrow B$  distinct indices
2:   for  $t = B - 1$  down to 0:  $r_t \leftarrow u \bmod (N - t)$ ;  $u \leftarrow \lfloor u / (N - t) \rfloor$ 
3:    $A \leftarrow [0, 1, \dots, N - 1]$ ; return  $[A.\text{pop}(r_t) \text{ for } t = 0 \dots B - 1]$ 
4: function PERMDecode( $\langle a_0, \dots, a_{B-1} \rangle, N$ )  $\triangleright$  indices  $\rightarrow$  integer
5:    $A \leftarrow [0, \dots, N - 1]$ ;  $u \leftarrow 0$ 
6:   for  $t = 0$  to  $B - 1$ :  $r \leftarrow A.\text{index}(a_t)$ ;  $u \leftarrow u(N - t) + r$ ;  $A.\text{pop}(r)$ 
7:   return  $u$   $\triangleright$  Equation (37)

```

Algorithm 4 ENCODE: message $\rightarrow$ ordered image sequence

```

1: Input: message  $m$ ; codebook  $\mathcal{C}$  (size  $N$ ); key  $\kappa$ ; flags  $perm$ , block  $B$ ,  $alias$ 
2:  $f \leftarrow \text{WRAP}(m, \kappa)$   $\triangleright$  Algorithm 2
3:  $v \leftarrow \text{INT}_{\text{BE}}(0x01 \parallel f)$   $\triangleright$  Equation (35)
4: if  $perm$  then
5:    $m' \leftarrow P(N, B)$ ;  $(e_0, \dots) \leftarrow \text{TOBASEN}(v, m')$ 
6:    $\text{idx} \leftarrow \text{CONCAT}_i \text{PERMEncode}((e_i + \mu_i) \bmod m', N, B)$   $\triangleright$  Equation (38)
7: else  $\text{idx} \leftarrow \text{TOBASEN}(v, N)$   $\triangleright$  Equation (36)
8: if  $alias$  then  $S \leftarrow [\text{CHOOSEALIAS}(j) \text{ for } j \in \text{idx}]$  else  $S \leftarrow [c_j \text{ for } j \in \text{idx}]$ 
9: return image sequence  $S$ 

```

padded, framed plaintext split into 16-byte blocks $t_1, t_2, \dots$, a random initialisation vector $c_0 = IV$, and the block cipher E_κ , the ciphertext blocks are produced by

$$c_i = E_\kappa(t_i \oplus c_{i-1}), \quad i = 1, 2, \dots, \quad (34)$$

and are recovered by the inverse recurrence $t_i = E_\kappa^{-1}(c_i) \oplus c_{i-1}$. The chaining in Equation (34) ensures that two identical plaintext blocks do not, in general, encrypt to identical ciphertext blocks, so that no block-level regularity of the payload survives into the emitted index stream. The inverse operation, UNWRAP, decrypts, separates the trailing four CRC bytes, and *rejects* the message if the recomputed check disagrees, thereby realising the guarantee of Equation (27). Cipher-block chaining is used so that identical plaintext blocks do not produce identical ciphertext blocks; the encryption is summarised in Algorithm 2.

3.2 LG-CISH coding channels

The framed bytes f of Equation (33) are turned into codebook indices by one of two lossless codes. In both cases f is first mapped to a single non-negative integer, with a 0x01 sentinel byte prepended so that leading zero bytes are preserved through the integer round-trip,

$$v = \text{INT}_{\text{BE}}(0x01 \parallel f). \quad (35)$$

Channel 1: base- N positional coding. The integer v is written in radix N , most-significant digit first, so that

$$v = \sum_{t=0}^{L-1} d_t N^t, \quad d_t \in \{0, \dots, N-1\}, \quad L = \lfloor \log_N v \rfloor + 1, \quad (36)$$

Algorithm 5 DECODE: image sequence $\rightarrow$ message

```

1: Input: received images  $\langle \tilde{y}_0, \dots, \tilde{y}_{T-1} \rangle$ ; codebook  $\mathbf{C}$ ; key  $\kappa$ ; flags  $perm, B$ 
2: for  $t = 0$  to  $T - 1$ :  $\hat{i}_t \leftarrow \arg \max_j s(e(\tilde{y}_t), \mathbf{C}_j)$   $\triangleright$  Equation (17)
3: if  $perm$  then
4:    $m' \leftarrow P(N, B)$ ; for each block  $i$ :  $e_i \leftarrow (\text{PERMDECODE}(\hat{i}_{[iB:iB+B]}, N) - \mu_i) \bmod m'$ 
5:    $v \leftarrow \text{FROMBASEN}((e_0, \dots), m')$ 
6: else  $v \leftarrow \text{FROMBASEN}(\hat{i}, N)$ 
7:  $f \leftarrow \text{STRIPSENTINEL}(\text{BYTESBE}(v))$   $\triangleright$  inverse of Equation (35)
8: return  $\text{UNWRAP}(f, \kappa)$   $\triangleright$  verify CRC, decrypt, decompress

```

and digit d_t selects image c_{d_t} . Because Equation (36) is a bijection, decoding simply reads the digits back; this channel attains the ceiling of Equation (16), at the cost of allowing an image to repeat. As a small illustration, suppose $N = 40$ and the integer to be sent is $v = 1\,000\,000$. Repeated division by 40 gives the digits $(9, 12, 20, 0)$, since $1\,000\,000 = 9 \cdot 40^3 + 12 \cdot 40^2 + 20 \cdot 40 + 0$, so the four images c_9, c_{12}, c_{20}, c_0 are emitted. The receiver, having recovered those indices, re-forms $9 \cdot 40^3 + 12 \cdot 40^2 + 20 \cdot 40 + 0 = 1\,000\,000$ and thereby recovers v exactly.

Channel 2: Lehmer-permutation coding. To obtain a cover in which no image repeats within a block, the integer v is instead written in radix $m = P(N, B)$ (Equation (23)), giving block values $(e_0, \dots, e_{L'-1})$. Each block value is expanded into B distinct images by a mixed-radix (Lehmer) code, in which the t -th image of a block is identified by its rank r_t among the still-available images (radix $N - t$):

$$u = \sum_{t=0}^{B-1} r_t \prod_{j=t+1}^{B-1} (N - j), \quad 0 \leq r_t < N - t. \quad (37)$$

A naïve Lehmer code maps a small leading block value to an almost-sorted image prefix, which is a visible artefact. It is removed by *whitening* each block value with a deterministic, shared keystream before expansion,

$$\tilde{e}_i = (e_i + \mu_i) \bmod m, \quad \mu_i = \text{SHA256}(\text{"perm-whiten-"}i) \bmod m, \quad (38)$$

which is invertible modulo m and preserves both losslessness and within-block distinctness. The two directions of the block code are given in Algorithm 3.

The mechanics of Equation (37) are best seen on a small example. Consider a codebook of $N = 4$ images and a block of $B = 4$, for which $P(4, 4) = 4! = 24$ orderings are available and a block therefore carries $\log_2 24 \approx 4.585$ bits. Suppose the whitened block value is $\tilde{e} = 17$. The ranks are read out by successive division with decreasing radix: $17 = 2 \cdot (3!) + 5$, then $5 = 2 \cdot (2!) + 1$, then $1 = 1 \cdot (1!) + 0$, giving the rank vector $(r_0, r_1, r_2, r_3) = (2, 2, 1, 0)$. Consuming these ranks from the available list $A = [0, 1, 2, 3]$ yields, in order, the indices 2, 3, 1, 0; that is, the block emits images c_2, c_3, c_1, c_0 , which are manifestly distinct. Decoding inverts the process: the rank of each emitted index among the remaining images is recomputed, and the mixed-radix sum of Equation (37) reconstructs $\tilde{e} = 17$, from which the whitening of Equation (38) is removed. The example makes concrete the two guarantees of the code — exact invertibility and within-block distinctness.

The choice between the two channels is a deployment decision rather than a correctness one, since both are lossless. Base- N coding is preferred when the overriding objective is to minimise the number of transmitted images, because it attains the ceiling of Equation (16); permutation coding

is preferred when the cover must survive human inspection as a plausible album, because the no-repeat guarantee removes the conspicuous artefact of the same image appearing several times in a short sequence. The block size B interpolates smoothly between the two regimes: a small block sacrifices little capacity relative to Equation (16) while still forbidding repeats within the block, whereas the full-block case $B = N$ maximises the no-repeat window at the largest capacity cost. In this way Equation (24) exposes a single, continuous control that a deployment can set according to whether compactness or plausibility is the more pressing requirement.

3.2.1 LG-CISH encoding

The complete sender procedure is assembled in Algorithm 4. The message is framed by Equations (32)–(33), mapped to the integer v by Equation (35), converted to indices by either Equation (36) or the whitened Lehmer code of Equation (38), and finally realised as images. When the optional alias mechanism of §3.3 is enabled, the image chosen for a given index is the alias that best improves cover plausibility; since every alias of an index decodes to the same index, this choice does not affect recovery.

For the decoder to invert the procedure, a small set of protocol parameters must be agreed in advance: whether compression and encryption were applied, which coding channel was used, and, for permutation coding, the block size B . These parameters are not secret — by Assumption 1.1 the warden may know them — and they occupy a negligible number of bits beside the payload; in a fixed deployment they are simply constants shared by both parties. With them fixed, Algorithm 4 is fully determined and its inverse, Algorithm 5, is unambiguous.

3.2.2 Lossless decoding

The receiver reverses the encoder exactly. Each received image is embedded with CLIP and matched to the nearest codebook entry by Equation (17), which recovers the transmitted index whenever the decoding margin of Equation (18) is positive; under Definition 3.1 and adequate transport stability this holds for every position, so the index stream is recovered without error. The recovered indices are then passed back through the inverse coding (Equation (36) or the un-whitened Lehmer inverse of Equations (37)–(38)) to reconstruct the integer v , the sentinel byte is stripped to obtain f , and UNWRAP verifies the CRC and inverts the encryption and compression. The procedure is given in Algorithm 5.

Proposition 3.1 (End-to-end losslessness). If every index is recovered correctly, i.e. $\gamma_t > 0$ for all t in Equation (18), then $\text{DECODE}(\text{ENCODE}(m)) = m$ exactly and $\text{BER} = 0$. This follows because each stage — Equation (32), Equation (33), Equation (35), Equation (36), and Equations (37)–(38) — is a bijection and is inverted in reverse order.

Computational cost. The asymmetry between the two directions of the scheme is worth noting. Encoding is dominated by big-integer arithmetic: the base- N conversion of Equation (36) costs $\mathcal{O}(L)$ big-integer operations for a sequence of L images, and the cryptographic frame is linear in the payload size; no neural network is evaluated on the sender side, so encoding is effectively instantaneous. Decoding is dominated by the feature map: each of the L received images is passed once through the CLIP encoder and then compared against the N codebook embeddings, giving a cost of

$$\mathcal{O}(L \cdot (C_\Phi + Nd)), \quad (39)$$

In Equation (39), C_Φ is the cost of a single CLIP forward pass and $d = 512$ is the embedding dimension. The nearest-neighbour term Nd is negligible beside the forward pass C_Φ , so decoding

time scales essentially linearly in the number of received images and is governed by the throughput of the image encoder. The memory footprint is a single $N \times d$ matrix of codebook embeddings, which is a few megabytes for the codebooks considered here and is independent of message length.

Behaviour beyond the operating range. It is instructive to consider what happens when a channel degradation exceeds the range that the codebook was provisioned to absorb. In that regime the margin of Equation (18) can turn negative at one or more positions, and by Equation (19) the message-level success probability falls sharply. Crucially, the scheme does not fail silently: because the recovered indices no longer invert to a byte string whose trailing CRC is consistent, the integrity check of §3.1.2 rejects the message. The failure is thereby made *visible*, and the standard recourse — retransmission, or a request for a less aggressive channel — is available. This behaviour is the practical realisation of the principle argued in §1.5, that a detected error is far preferable to an undetected one, and it is the reason the CRC is retained even when encryption is disabled.

3.3 Plausibility aliases

The undetectability argument of §1.6 concerns the *marginal* distribution of each transmitted image, which is naturally innocent. A separate, weaker concern is whether the *sequence* of images looks like a plausible personal album to a human observer. To address this without touching losslessness, each codebook slot j is optionally equipped with a small set of visually interchangeable candidate images, called aliases, that have been verified by a vision-capable large language model to depict the same subject as c_j . All aliases of slot j map to the same index j , so the decoder is entirely unaffected. During encoding, whenever index j is to be emitted, the alias that best improves the coherence of the running sequence is selected. Because the choice among aliases is invisible to the index-recovery rule of Equation (17), cover plausibility is improved as a free, orthogonal degree of freedom, and Proposition 3.1 continues to hold.

In more detail, the alias set for each slot is prepared offline by a generate-and-verify loop. Candidate images that plausibly depict the same subject as c_j are proposed — either retrieved from the database or synthesised — and each candidate is then submitted, together with the anchor image c_j , to a vision-capable large language model that is asked to confirm that the two depict the same subject and to score their interchangeability. Only candidates whose confirmation passes a fixed threshold are retained, so that the alias relation is conservative by construction. Because this verification is performed once, offline, it adds nothing to the per-message encoding cost; and because the resulting aliases all share the slot index j , they remain invisible to the recovery rule of Equation (17). It is stressed that aliases affect only the human plausibility of the *ordering* of the cover, not the distributional undetectability of each individual image, which is already guaranteed by Equation (14).

4 Experimental Results and Performance Evaluation

The evaluation is organised around the three quantities that any steganographic system must trade against one another — imperceptibility, capacity, and robustness — followed by benchmark comparison, reconstruction and matching metrics, an ablation study, and steganalysis resistance. Throughout, the codebook consists of $N=40$ mutually distinct natural images (Section ??), each normalised to a fixed 512×512 canvas, giving $\log_2 40 \approx 5.322$ bits per image. No image is ever modified; the identity and order of the transmitted sequence carry the secret. The full suite is

reproducible from a single script.¹

Metrics. For a message encoded into L images with intended digits d_t and recovered digits $\hat{d}_t$, and B payload bits $p_b/\hat{p}_b$, the symbol (hash-matching) error rate and the bit error rate are

$$\text{SER} = \frac{1}{L} \sum_{t=1}^L \mathbb{1}[\hat{d}_t \neq d_t], \quad \text{BER} = \frac{1}{B} \sum_{b=1}^B \mathbb{1}[\hat{p}_b \neq p_b]. \quad (40)$$

Hash-matching accuracy is $1 - \text{SER}$, and reconstruction is *exact* if and only if $\text{SER} = 0$, which, by the bijectivity of the base- N coding, implies $\text{BER} = 0$.

5 Conclusion

LG-CISH is a lossless coverless image-steganography scheme that pairs CLIP semantic hashing with base- N positional coding, with an optional distinct-image permutation mode and an LLM-guided alias layer for cover shaping. It transmits only unmodified natural images, achieving bit-exact reconstruction ($\text{BER} = 0$), 100% robustness across twenty of twenty-one channel attacks (graceful degradation only under a $10\times$ downscale), chance-level steganalysis detectability (47.5% vs. 97.5% for LSB), and a statistically significant geometric-robustness advantage over perceptual hashing ($p=7.8 \times 10^{-3}$). Its limitations are stated honestly: capacity is database-bound and is addressed by scaling the codebook, while cover plausibility is theme-bound and is addressed by a coherent database. Taken together, the results indicate that a coverless channel built on semantic hashing can attain, simultaneously, the imperceptibility and robustness that pixel-domain methods must trade against one another.

5.1 Future Scope

Several directions are left for future work. *First*, the codebook can be scaled from the present $N=40$ to $N=10^3$ – 10^4 images, which raises the per-image capacity to 10–13 bits and shortens the transmitted sequence, subject to maintaining the separation margin as the embedding space becomes more densely populated. *Second*, the plausibility of the cover *ordering* can be strengthened by tightening the large-language-model alias and re-selection loop around a thematically coherent database, so that the transmitted sequence reads as a natural personal album rather than a set of unrelated images. *Third*, although undetectability holds by construction against pixel-domain detectors, a dedicated adversarial study against end-to-end convolutional steganalysers (SRNet, XuNet) trained specifically on this channel would further substantiate the security claim. *Fourth*, the distinct-image permutation and a dual-channel identity–ordering code can be combined to recover part of the capacity conceded to the coverless paradigm. *Finally*, the framework can be extended beyond still images to coverless video and audio channels, and validated in a real-world deployment over public messaging platforms, together with a study of database versioning and key management for practical multi-party use.

Acknowledgements

The authors acknowledge the use of benchmark images from the UCID database, the Kodak Lossless True Color Image Suite, and the USC-SIPI Image Database. The USC-SIPI images

¹`python evaluation/generate_all.py`

were obtained from Volume 3, including images available under the Miscellaneous category at <https://sipi.usc.edu/database/index.php?volume=misc&image=10#top>.

Ethics Declaration

This research did not involve studies with human participants or animals and did not use sensitive personal data. Therefore, ethical approval and informed consent were not required.

Data Availability

The datasets, implementation, evaluation scripts, and the released 40-image LG-CISH codebook supporting the findings of this research are available in a publicly accessible GitHub repository:

GitHub Repository: <https://github.com/Circuit-Overtime/LG-CISH>

Funding Declaration

The authors declare that no funding, grants, or other financial support were received for the conduct of this research or the preparation of this manuscript.

References

- [1] N. Provos and P. Honeyman, “Hide and seek: An introduction to steganography,” *IEEE Security & Privacy*, vol. 1, no. 3, pp. 32–44, 2003.
- [2] G. J. Simmons, “The prisoners’ problem and the subliminal channel,” in *Advances in Cryptology (CRYPTO 1983)*, Springer, pp. 51–67, 1984.
- [3] C.-K. Chan and L. M. Cheng, “Hiding data in images by simple LSB substitution,” *Pattern Recognition*, vol. 37, no. 3, pp. 469–474, 2004.
- [4] A. Westfeld, “F5—a steganographic algorithm,” in *Information Hiding (IH 2001)*, LNCS 2137, pp. 289–302, 2001.
- [5] C. Cachin, “An information-theoretic model for steganography,” in *Information Hiding (IH 1998)*, LNCS 1525, pp. 306–318, 1998.
- [6] A. D. Ker, T. Pevny, J. Kodovsky, and J. Fridrich, “The square root law of steganographic capacity,” in *Proc. 10th ACM Workshop on Multimedia and Security*, pp. 107–116, 2008.
- [7] G. Xu, H.-Z. Wu, and Y.-Q. Shi, “Structural design of convolutional neural networks for steganalysis,” *IEEE Signal Processing Letters*, vol. 23, no. 5, pp. 708–712, 2016.
- [8] M. Boroumand, M. Chen, and J. Fridrich, “Deep residual network for steganalysis of digital images,” *IEEE Trans. Information Forensics and Security*, vol. 14, no. 5, pp. 1181–1193, 2019.

- [9] Z. Zhou, H. Sun, R. Harit, X. Chen, and X. Sun, “Coverless image steganography without embedding,” in *Int. Conf. Cloud Computing and Security (ICCCS)*, LNCS 9483, pp. 123–132, 2015.
- [10] X. Xiang, Y. Tan, J. Qin, and Y. Tan, “Advancements and challenges in coverless image steganography: A survey,” *Signal Processing*, vol. 228, 2024.
- [11] A. Radford, J. W. Kim, C. Hallacy, et al., “Learning transferable visual models from natural language supervision,” in *Proc. 38th Int. Conf. Machine Learning (ICML)*, PMLR 139, pp. 8748–8763, 2021.
- [12] N. Subramanian, O. Elharrouss, S. Al-Maadeed, and A. Bouridane, “Image steganography: A review of the recent advances,” *IEEE Access*, vol. 9, pp. 23409–23423, 2021.
- [13] M. H. Kombrink, Z. J. M. H. Geradts, and M. Worring, “Image steganography approaches and their detection strategies: A survey,” *ACM Computing Surveys*, vol. 57, no. 2, art. 33, 2024.
- [14] J. Qin, Y. Luo, X. Xiang, Y. Tan, and H. Huang, “Coverless image steganography: A survey,” *IEEE Access*, vol. 7, pp. 171372–171394, 2019.
- [15] X. Li, J. Yu, J. Cheng, Z. Li, and C. Bian, “Semantic preservation-based hash code generation for fine-grained image retrieval,” *Expert Systems with Applications*, vol. 271, art. 126668, 2025.
- [16] I. J. Kadhim, P. Premaratne, P. J. Vial, and B. Halloran, “Comprehensive survey of image steganography: Techniques, evaluations, and trends in future research,” *Neurocomputing*, vol. 335, pp. 299–326, 2019.
- [17] P. C. Mandal, I. Mukherjee, and B. N. Chatterji, “Digital image steganography: A literature survey,” *Information Sciences*, vol. 609, pp. 1451–1488, 2022.
- [18] A. Westfeld and A. Pfitzmann, “Attacks on steganographic systems,” in *Information Hiding (IH 1999)*, LNCS 1768, pp. 61–76, 2000.
- [19] J. Fridrich, M. Goljan, and R. Du, “Detecting LSB steganography in color and gray-scale images,” *IEEE MultiMedia*, vol. 8, no. 4, pp. 22–28, 2001.
- [20] H. Kheddar, M. Hemis, Y. Himeur, D. Megias, and A. Amira, “Deep learning for steganalysis of diverse data types: A review of methods, taxonomy, challenges and future directions,” *Neurocomputing*, vol. 581, art. 127528, 2024.
- [21] R. Apau, M. Asante, F. Twum, J. B. Hayfron-Acquah, and K. O. Peasah, “Image steganography techniques for resisting statistical steganalysis attacks: A systematic literature review,” *PLOS ONE*, vol. 19, no. 9, art. e0308807, 2024.
- [22] L. Meng, X. Jiang, and T. Sun, “A review of coverless steganography,” *Neurocomputing*, vol. 566, art. 126945, 2024.

- [23] L. Zou, J. Li, W. Wan, Q. M. J. Wu, and J. Sun, “Robust coverless image steganography based on neglected coverless image dataset construction,” *IEEE Trans. Multimedia*, vol. 25, pp. 5552–5564, 2023.
- [24] L. Meng, X. Jiang, Z. Zhang, Z. Li, and T. Sun, “A robust coverless image steganography based on an end-to-end hash generation model,” *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 7, pp. 3542–3558, 2023.
- [25] F. Li, C. Liu, Z. Dong, Z. Sun, and W. Qian, “A robust coverless image steganography algorithm based on image retrieval with SURF features,” *Security and Communication Networks*, art. 5034640, 2024.
- [26] L. Meng, F. Li, X. Jiang, and Q. Xu, “A universal framework for improving the robustness of coverless image steganography based on image restoration,” *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 1, pp. 922–937, 2025.
- [27] Q. Zhang and F. Huang, “Robust generative steganography based on image mapping,” *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 13543–13555, 2024.
- [28] Y. Yang, Z. Liu, J. Jia, Z. Gao, Y. Li, W. Sun, X. Liu, and G. Zhai, “DiffStega: Towards universal training-free coverless image steganography with diffusion models,” in *Proc. 33rd Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2024.
- [29] B. Song, P. Wei, S. Wu, Y. Lin, and W. Zhou, “A survey on deep-learning-based image steganography,” *Expert Systems with Applications*, vol. 254, art. 124390, 2024.
- [30] S. Baluja, “Hiding images in plain sight: Deep steganography,” in *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 2069–2079, 2017.
- [31] M. Tancik, B. Mildenhall, and R. Ng, “StegaStamp: Invisible hyperlinks in physical photographs,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 2117–2126, 2020.
- [32] Z. Hu, Y.-M. Cheung, M. Li, W. Lan, D. Zhang, and Q. Liu, “Joint semantic preserving sparse hashing for cross-modal retrieval,” *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 4, pp. 2989–3002, 2024.
- [33] J. Daemen and V. Rijmen, *The Design of Rijndael: The Advanced Encryption Standard (AES)*, 2nd ed. Springer, 2020.
- [34] G. Schaefer and M. Stich, “UCID: An uncompressed color image database,” in *Proc. SPIE 5307, Storage and Retrieval Methods and Applications for Multimedia*, pp. 472–480, 2003.
- [35] R. Franzen, “Kodak lossless true color image suite (PhotoCD PCD0992),” Eastman Kodak Company, 1999.
- [36] A. G. Weber, “The USC-SIPI image database,” Signal and Image Processing Institute, Univ. of Southern California, Tech. Rep., 1997.

- [37] E. Agustsson and R. Timofte, “NTIRE 2017 challenge on single image super-resolution: Dataset and study,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 126–135, 2017.
- [38] A. Bhattacharya, S. Dutta, S. Biswas, P. Sarkar, and H. Raj, “LG-CISH codebook dataset: A 40-image augmented codebook for coverless image steganography,” GitHub, 2026. [Online]. Available: <https://github.com/Circuit-Overtime/LG-CISH>
- [39] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, “Image quality assessment: From error visibility to structural similarity,” *IEEE Trans. Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.